The distributive law

1 A student incorrectly used the distributive property and wrote $6(7x+2)=42x+2$. Explain how to correct the error.

2 Expand the following expressions:

a $5(s+7)$
b $9(5+w)$
c $3(r-6)$
d $(y+8)\times 7$
e $-8(t+10)$
f $-8(c-5)$
g $-(y+2)$
h $4(4s+5)$
i $-10(8+3r)$
j $-8(-2t-3)$
k $p(q+4)$
l $9(u-v)$
m $-(2m-3n)$
n $-r(11s+13t)$
o $4(2r+3s+5t)$
p $6(4m+3n+5p)$

3 Expand the following expressions:

a $6m(2n+3)$
b $9\left(v^2+6\right)$
c $r(r+7)$
d $6m(4m+7)$
e $-6a(2a+9)$
f $4wy(y+w)$
g $4p(4q-3+6r)$
h $2f(f-7g-6h)$
i $-10\left(y^2+7\right)$
j $-\left(9-s^3\right)$
k $r\left(s^2+8\right)$
l $-10\left(3+8u^3\right)$
m $4s^5\left(4t^4-9\right)$
n $6p^3\left(2q^5-9\right)$
o $9f(f-4g-6h)$
p $3uv(v+u)$

4 Expand and simplify the following expressions:

a $4(x+8)-2$
b $7+5(3x+4)$
c $5+3(x+4)$
d $8(3x+4)-6x$
e $34+9(4x-5)$
f $5x-8(2x-3)$
g $9x-5(2x+3)$
h $-8(5x-7)-9$
i $4y+5+6(y-9)$
j $4(5(x-3)+2)$

5 Expand and simplify the following expressions:

a $x(x+4)+8(x+9)$
b $8y^2-4y+7y(y-5)$
c $3(6x-5y)-2(9y+4x)$
d $7(y+9)+5(y+6)$
e $10(x-4)+3(x+8)$
f $5(y+7)-(y-9)$
g $-8(x+4)-3(x-2)$
h $11(a+4)-6(4-a)$
i $4u(6u+9v)-2u(8v+5u)$
j $9w(8w+6z)-5w(4z+2w)$

Applications

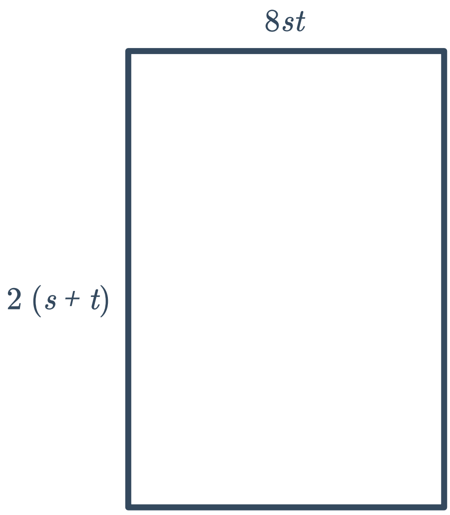
6 A solar energy company offers a range of solar panels, each with a power rating of

$(m+8)$ kW, where m is the panel model number. Write an expression in terms of m for the total amount of power produced by 11 solar panels.

7 The representatives of 8 countries are attending a global summit, and each country is able to send $(2r-3)$ representatives. Write an expression for the total number of representatives attending the summit.

8 A forklift operator has enough time to make 5 trips between the two ends of a warehouse. A forklift can carry $(2+3n^4)$ boxes at a time. Write an expression for the total number of boxes that can be carried across the warehouse.

9 Consider the rectangle shown:
Find an expression for the area of the rectangle in expanded form.



10 A rectangle is 8 m longer than it is width. Its width is represented by the pronumeral u. Form an expression for the area of the rectangle in expanded form.

Factorisation

11 State whether the following are factors of $33wy$:

a 11 b $66wy$ c yw d $3w$
e yw^2 f 33

12 Find the highest common factor of the following groups of numbers:

a 2 and $6x$ b 3 and $15x$
c $6x$ and 21 d 10 and $16u$
e $35x$ and 14 f $4x$ and $16xy$
g 15 and $5x$ h $24m$, $12n$ and $18q$

13 Complete the factorisation of the following expressions:

a $2a+16=\square\left(a+8\right)$

b $5y-30=\square\left(y-6\right)$

c $24n+15=\square\left(8n+5\right)$

d $16a-6=\square\left(8a-3\right)$

e $6v+18=6\left(\square\right)$

f $9a-36=9\left(\square\right)$

g $24a+9=3\left(\square\right)$

h $6w-27=3\left(\square\right)$

i $4r+12=\square\left(r+3\right)$

j $5v-30=\square\left(v-6\right)$

k $18c-24=\square\left(3c-4\right)$

l $2r+14=2\left(\square\right)$

m $5t-10=5\left(\square\right)$

n $54u+12=6\left(\square\right)$

14 Factorise the following expressions:

a $6v+30$

b $6a-48$

c $45t-40$

d $2m+10n$

e $54-60fg$

f $8g-72h$

g $9f+15g$

h $3w+15$

i $2c-18$

j $40c-35$

k $6w+30y$

l $14y+4w$

m $4a-14b$

n $32-36jk$

o $8jkt-18$

p $3fg-10fh$

q $3rtw-8rw$

r $4rt-6r^2t$

s $16wy-36w^2y$

15 Complete the following factorisation:

$$2w-10z+8y=2\left(\square-\square+\square\right)$$

16 Factorise the following completely:

a $8g+28h-20$

b $4w+4z+4y$

c $4u-12v+20w$

d $10m-25n-15p$

e $3m-6n+12p$

f $10g-35h+15$

g $zw-zy-zk$

h $60r-25rt-35r^2t$

i $12m-32mn-20m^2n$

j $16z-32yz-24y^2z$

k $25r^2t-15rt^2-45r^2t^2$

Factorisation of negatives

17 Factor out the negative coefficient of the following expressions:

a $-4r-3s$

b $-7r-7s$

c $-2-9r$

d $-8x-5y$

18 Factorise the following expressions:

a $-8b-16$	b $-18a+16$	c $-80v-30$	d $-36-6x$
e $-25u-35v$	f $-35m+49n$	g $-8pqr-6pr$	h $-8w^2+3w^2y$
i $-24j^2-27j^2k$	j $-8m^2+9mn$	k $-10u^2v+9uv^2$	l $-21j^2-36j^2k$
m $-2s-10$	n $-8v^2+56$	o $-12s+10$	p $-28b-21$
q $-63r^2+28$	r $-4m-32n$	s $-49y-21w$	

19 Factorise the following completely:

a $-18p-9q-45r$	b $-10r-25s+20t$
c $-20u-25v+45w$	d $-8w-18y-10z$